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Correct Answer

Next C

Practice Exam - 1

Question 50 of 117



A patient presents with new-onset type 2 diabetes with complaints of increased thirst and urination at night. He has a past medical history of HFpEF (LVEF of 55%) hypertension, STEMI, dyslipidemia, obesity, and chronic kidney disease. He has an eGFR of 24 L/min/1.73 m, hemoglobin A1c of 7.8%, and BMI is 36 kg/m². Pertinent labs consist of a potassium 4.0 mEq/L and a serum creatinine of 2.8 mg/dl, which is his baseline. He is currently stabilized on sacubitril/valsartan, carvedilol, spironolactone, aspirin, atorvastatin and has not been able to tolerate dapagliflozin due to multiple urinary tract infections.

Which of the following would be the best therapeutic option to manage this patient's type 2 diabetes?

A. Insulin

 B. Exenatide

 C. Semaglutide

D. Alogliptin

Commentary

References

Testing Point: The board-certified AHFTC specialist should select appropriate diabetes therapy in a patient with HFpEF, obesity, ASCVD, and chronic kidney disease who cannot tolerate SGLT2 inhibitor therapy.

Correct Answer: C. Semaglutide

Rationale: Based on the 2020 American Diabetes Association (ADA) guidelines, choice of therapeutic interventions depends upon presence of comorbidities such as history of heart failure, chronic kidney disease (CKD), and cardiovascular (CV) disease; risk factors for atherosclerotic cardiovascular disease (ASCVD); and evaluation of renal function. Based on this patient's established ASCVD, HF with reduced ejection fraction, and CKD, the ADA would recommend a SGLT2-inhibitor first and then consider a glucagon-like peptide (GLP)-1 receptor agonist with CVD benefit. However, as a therapeutic class, blood glucose reduction is significantly blunted with the SGLT2-inhibitors in eGFR < 45 ml/min/m², and this patient did not tolerate dapagliflozin due to multiple urinary tract infections. He

has an eGFR of 24 ml/min/m².

Both Answer B (exenatide) and Answer C (semaglutide) are GLP-1 agonists, however only liraglutide, semaglutide, and dulaglutide have demonstrated both CV and renal protective benefits, making Answer B incorrect and Answer C correct.

In the SUSTAIN trial, semaglutide (Answer C) was associated with a 26% reduction in CV death, nonfatal MI, or nonfatal stroke (HR: 0.74, 95% CI: 0.5-0.95), while exenatide in the EXSCEL study suggested less or no CV benefit (HR: 0.91, 95% CI: 0.83-1.00). Additionally, semaglutide is approved for eGFR of > 15 ml/min/m² while exenatide is approved at an eGFR > 30 ml/min/m². While effective at lowering blood sugar acutely, insulin is not recommended first line for long term management, as it has been associated with an increased risk of major cardiac events, worsening edema in heart failure, and does not preserve kidney function, making Answer A incorrect.

There is now a clinical trial database showing benefit of semaglutide in patients with HFpEF and obesity (STEP-HFpEF program).

Finally, alogliptin (Answer D) is a dipeptidyl peptidase-4 (DPP-4) inhibitor which has no additional cardiac or renal proactive benefits and should be avoided in HF due to worsening of symptoms, making Answer D incorrect.

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